Histograms

Exam code: 9MA0

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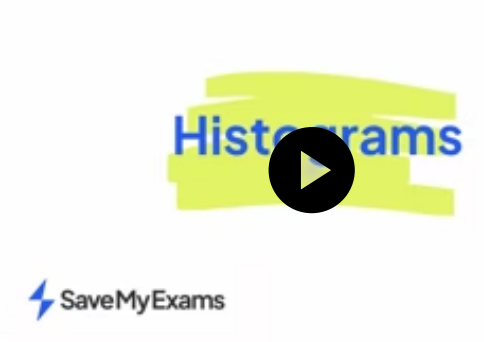

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Histograms

What is a histogram?

- A histogram is similar to a bar chart but with some **key differences**
  - A histogram is for displaying **grouped continuous data** whereas a bar chart is for **discrete** or **qualitative data**
  - There will never be any gaps between the bars of adjacent groups in a histogram
  - Whilst in a bar chart the frequency is read from the **height** of the bar, in a histogram the height of the bar is the **frequency density**
- On a histogram **frequency density** is plotted on the y – axis
  - This allows a histogram to be plotted for **unequal class intervals**
  - It is particularly useful if data is spread out at either or both ends
- The **area** of each bar on a histogram will be **proportional** to the frequency in that class

What are the key features of a histogram?

- You will not be asked to draw a histogram but you may have to add information to one so you should make sure

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

next class
    - If there are gaps you will need to close them by changing the boundaries before carrying out any calculations
        - Consider whether the values are rounded or truncated before closing the gaps

- **Step 2.** Find the **class width** of each group by subtracting the lower boundary from the upper boundary
- **Step 3**. Calculate the **frequency density** for each group using the formula:

$$\text{frequency density} = k \times \frac{\text{frequency}}{\text{class width}}$$

- **Step 4.** The histogram will be drawn with the data values on the x – axis and frequency density on the y – axis
    - Remember that the scale on both axes must be even, although the class widths may be uneven
        - Both axes should be clearly labelled and units included on the x – axis
    - Most often, the bars will have different widths
- Occasionally you will be asked to add a **frequency polygon** to the histogram
    - This is done by joining up the midpoints at the top of each bar
    - You should not join up the first or last midpoint to the x – axis (it is not really a polygon!)

How do I interpret a histogram?

- It is important to remember that the y – axis does not tell us the frequency of each bar in the histogram
- The **area** of the bar gives information about the frequency
    - Most of the time, the frequency will be the area of the bar and is found by multiplying the class width by the frequency density
    - Occasionally, the frequency will be proportional to the area of the bar
        - Frequency = $k \times \text{area}$
        - In these cases more information will be given to help you find the value of k
- You may be asked to find the frequency of part of a bar within a histogram
    - Find the area of that section of the bar using any information you have already found out
    - You will need to have found the value of k first

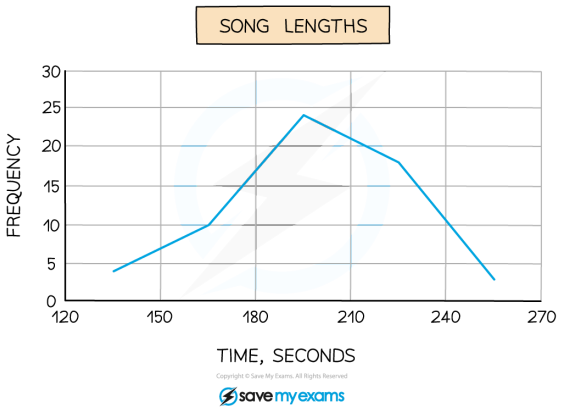
What are frequency polygons?

- **Frequency polygons** are a very simple way of showing frequencies/frequency densities for **continuous**, **grouped** data and give a quick guide to how frequencies change from one class to the next

What do I need to know about frequency polygons?

- Unless one of the frequencies/densities is 0 do not join the frequency polygon to the x-axis, and do not join the first point to the last one

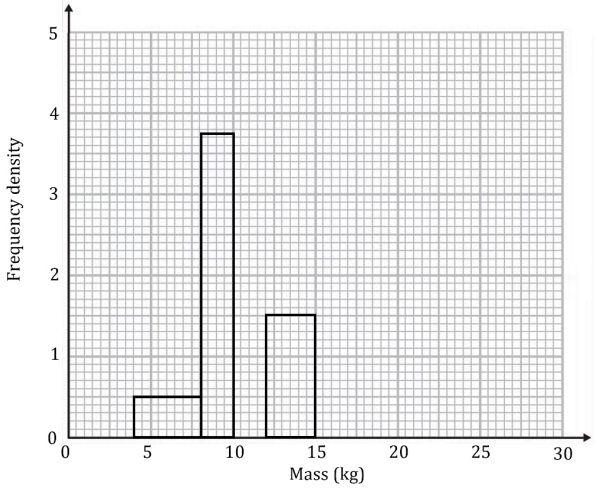
- The result is not actually a polygon but more of an open one that 'floats' in mid-air!
- You may be asked to draw a frequency polygon and/or use it to make comments and compare data



Worked Example

The table below and its corresponding histogram show the mass, in kg, of some new born bottlenose dolphins.

Mass, m kg	Frequency
$4 \le m < 8$	4
$8 \le m < 10$	15
$10 \le m < 12$	19
$12 \le m < 15$	
$15 \le m < 30$	6



(ii) Estimate the number of dolphins whose weight is greater than 13 kg.

Answer:

Examiner Tips and Tricks

- Remember that the area of a bar in a histogram is not always the frequency itself but could be proportional to the frequency. Look carefully at the scales on the axes, it will rarely be a simple 1 unit to 1 square.

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